

Differencing a Time Series

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Introduction

Differencing is the simplest tool for making a non-stationary series stationary: replace each observation by its difference from the previous one. First differences remove linear trends; seasonal differences remove fixed-period seasonality; combined regular and seasonal differencing handles both. The order of differencing d in $\text{ARIMA}(p, d, q)$ is exactly the count of first differences applied; the seasonal differencing order D in $\text{SARIMA}(p, d, q)(P, D, Q)_s$ counts seasonal differences.

Prerequisites

A working understanding of stationarity, AR models, and the relationship between trend, seasonality, and the spectrum of a series.

Theory

Standard differencing operations:

- **First differences:** $\Delta y_t = y_t - y_{t-1}$. Removes linear (level-shift) non-stationarity.
- **Second differences:** $\Delta^2 y_t = \Delta y_t - \Delta y_{t-1}$. Removes quadratic trend. Rarely needed in practice; second differences typically over-difference and inflate residual variance.
- **Seasonal differences** at period s : $\Delta_s y_t = y_t - y_{t-s}$. Removes constant seasonal patterns at frequency $1/s$.
- **Combined regular and seasonal:** $\Delta \Delta_s y_t = (y_t - y_{t-1}) - (y_{t-s} - y_{t-s-1})$.

Over-differencing introduces unnecessary moving-average structure (the differenced series has a strong negative lag-1 autocorrelation); always use the minimum differencing order needed for stationarity. KPSS- or ADF-based automated selection (`forecast::ndiffs` and `nsdiffs`) chooses the required orders.

Assumptions

The non-stationarity is of the kind that differencing addresses (stochastic trend or fixed-period seasonality); deterministic polynomial trends are better handled by detrending regression.

R Implementation

```
library(forecast)

x <- AirPassengers
plot(x)

# First difference
plot(diff(x))

# Seasonal difference (lag = 12 for monthly)
plot(diff(x, lag = 12))
```

```
# Both
plot(diff(diff(x, lag = 12)))

# Automatic: number of differences needed
ndiffs(log(x))      # regular
nsdiffs(log(x))     # seasonal
```

Output & Results

`diff(x, lag = s, differences = d)` produces $\Delta_s^d y$. `ndiffs()` returns the required regular differencing order based on KPSS by default; `nsdiffs()` returns the seasonal differencing order. For the log-AirPassengers series, both regular and seasonal differencing once each produces a stationary residual.

Interpretation

A reporting sentence: “Monthly air-passenger counts on the log scale required one regular difference and one seasonal difference at lag 12 to achieve stationarity (KPSS $p > 0.10$ on the doubly-differenced series); the resulting series motivates an $\text{ARIMA}(0, 1, 1)(0, 1, 1)_{12}$ model on the log scale, equivalent to a multiplicative seasonal ARIMA on the original scale.” Always confirm stationarity after differencing.

Practical Tips

- Log-transform first for series with variance that grows with the level; the log scale is then additive and the differencing is well-behaved.
- $\text{ARIMA}(p, d, q)$ encodes the differencing order d ; the corresponding `Arima()` call automatically applies the difference and works with the original series for prediction.
- `forecast::ndiffs()` and `nsdiffs()` automate the choice of regular and seasonal differencing orders; useful as a starting point for `auto.arima`.
- Over-differencing inflates residual variance and introduces a strong negative MA(1) coefficient; if you see θ_1 near -0.95 in a differenced model, consider whether the differencing was excessive.
- Always confirm stationarity after differencing with ADF and KPSS; if either test still indicates non-stationarity, examine the data for structural breaks or non-linear trends.
- For deterministic polynomial trends, detrending regression is more efficient than differencing; ARIMA’s stochastic-trend interpretation differs.

R Packages Used

Base R `diff()` for differencing operations; `forecast::ndiffs()`, `nsdiffs()`, and `auto.arima()` for automated differencing decisions; `feasts::unitroot_ndiffs()` and `unitroot_nsdiffs()` for the modern tidyverts equivalents; `tseries::adf.test()` and `kpss.test()` for confirming stationarity after differencing.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Time series basics